

Complex Solution Sets

Answer Key

Algebra 2

Quick Answers

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|---|--|
| 1. $7i, -7i$ | 8. $1, -1, 2i, -2i$ |
| 2. $5i, -5i$ | 9. the value after substituting $= 0$, the complete solution set $= 1 + 2i, 1 - 2i$ |
| 3. $2 + 3i, 2 - 3i$ | 10. $2, 3i, -3i$ |
| 4. $3, -3, 3i, -3i$ | 11. $-3 + 5i, -3 - 5i$ |
| 5. the complete solution set $= 4i, -4i$, what was missing and why $=$ — | 12. the quadratic in standard form $= x^2 - 8x + 17$, the justification $=$ — |
| 6. $-1 + 2i, -1 - 2i$ | |
| 7. $5 + 3i, 5 - 3i$ | |
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What is verified

13 of 15 answers machine-checked against SymPy, across 12 problems.

— marks an answer only you can judge — problems 5, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 2

HSN-CN.C.7 – problems 1, 2, 3, 5, 6, 7, 9, 11

HSN-CN.C.9 – problems 4, 8, 10, 12

Difficulty 1–5 of 5 across 15 tagged checks.

1. Solve $x^2 + 49 = 0$.

Solution: Isolate the square: $x^2 = -49$. A square root of a negative number is imaginary, and there are two of them.

$x = 7i$ or $x = -7i$

2. Solve $3x^2 + 75 = 0$.

Solution: Divide by 3 first: $x^2 + 25 = 0$, so $x^2 = -25$. Both square roots count.

$x = 5i$ or $x = -5i$

3. Solve $x^2 - 4x + 13 = 0$.

Solution: The discriminant is $(-4)^2 - 4(1)(13) = 16 - 52 = -36$, which is negative, so both solutions are non-real.

$$x = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2}$$

Dividing both parts by 2 gives the standard form.

$x = 2 \pm 3i$

4. Solve $x^4 - 81 = 0$, listing all four solutions.

Solution: Factor as a difference of squares, twice.

$$x^4 - 81 = (x^2 - 9)(x^2 + 9) = 0$$

The first factor gives $x^2 = 9$; the second gives $x^2 = -9$, which is where the two non-real solutions live. Stopping after the real pair is the classic incomplete answer.

$x = 3, -3, 3i, -3i$

5. A student solved $x^2 + 16 = 0$ and reported only $4i$. (a) Complete solution set? (b) What is missing and why?

Solution (a): $x^2 = -16$, and a square root has two values.

$x = 4i$ or $x = -4i$

Part (b) — instructor-judged. A full-credit explanation says the reported set is incomplete because $(-4i)^2 = 16i^2 = -16$ as well, so $-4i$ is a solution too, and a second-degree equation has exactly two solutions counted with multiplicity. A student who substitutes $-4i$ back and shows it gives zero has demonstrated it. Adding the missing value with no reason for why a second one must exist does not earn the mark.

6. Solve $2x^2 + 4x + 10 = 0$.

Solution: Divide through by 2 first — it makes the discriminant much easier: $x^2 + 2x + 5 = 0$. Now the discriminant is $4 - 20 = -16$, so

$$x = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2}.$$

Dividing both parts by 2:

$x = -1 \pm 2i$

7. Solve $(x - 5)^2 + 9 = 0$.

Solution: The square is already isolated, so no formula is needed: $(x - 5)^2 = -9$, hence $x - 5 = \pm 3i$. Adding 5 to both sides gives the standard form.

$x = 5 \pm 3i$

8. Solve $x^4 + 3x^2 - 4 = 0$, listing all four solutions.

Solution: This is a quadratic in x^2 : factor it as $(x^2 + 4)(x^2 - 1) = 0$. The second factor gives $x^2 = 1$; the first gives $x^2 = -4$.

$x = 1, -1, 2i, -2i$

9. $x^2 - 2x + 5 = 0$. (a) Substitute $x = 1 + 2i$. (b) Write the complete solution set.

Solution (a): $(1 + 2i)^2 = 1 + 4i + 4i^2 = -3 + 4i$, and $-2(1 + 2i) = -2 - 4i$. Adding all three terms: $(-3 + 4i) + (-2 - 4i) + 5$. 0

Solution (b): Because the left side is zero there, $1 + 2i$ is a solution; the coefficients are real, so its conjugate is the other one. (The quadratic formula confirms it: the discriminant is $4 - 20 = -16$.) $x = 1 \pm 2i$

10. Find all zeros of $f(x) = x^3 - 2x^2 + 9x - 18$.

Solution: Group in pairs and factor:

$$\begin{aligned}x^3 - 2x^2 + 9x - 18 &= x^2(x - 2) + 9(x - 4) \\&= (x - 2)(x^2 + 9)\end{aligned}$$

The linear factor gives one real zero; $x^2 = -9$ gives the conjugate pair. A cubic has three zeros, and here only one of them is real. $x = 2, 3i, -3i$

11. Solve $x^2 + 6x + 34 = 0$.

Solution: The discriminant is $36 - 136 = -100$, so

$$x = \frac{-6 \pm \sqrt{-100}}{2} = \frac{-6 \pm 10i}{2}.$$

Both parts of the numerator are divided by 2 — dividing only the radical part is the usual slip here. $x = -3 \pm 5i$

12. A monic quadratic with real coefficients has $4 - i$ as a solution. (a) Write it in standard form. (b) Justify why $4 + i$ is the other solution.

Solution (a): The two factors are $x - (4 - i)$ and $x - (4 + i)$. Multiplying them, the i terms in the middle cancel:

$$\begin{aligned}(x - (4 - i))(x - (4 + i)) &= x^2 - 8x + (4 - i)(4 + i) \\&= x^2 - 8x + (16 + 1)\end{aligned}$$

$x^2 - 8x + 17$

Part (b) — instructor-judged. A full-credit justification says that a polynomial with real coefficients has its non-real solutions in conjugate pairs, so the partner of $4 - i$ is $4 + i$. A justification by construction earns full credit too: the imaginary parts cancel only when the second factor is the conjugate, which is what makes the middle coefficient the real number -8 . Asserting the pair rule with no reason attached does not.